THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT NO.1

OBJECT ORIENTED PROGRAMMING

Submitted by-

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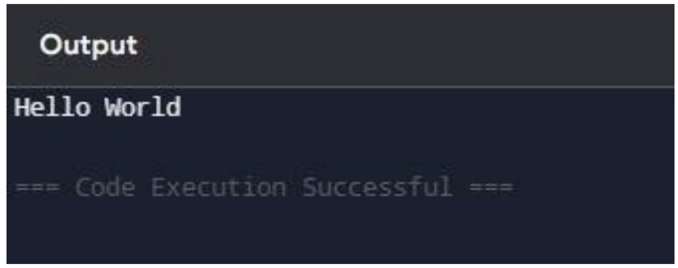
Q1. Write a program (WAP) to display "Hello World" on console display.

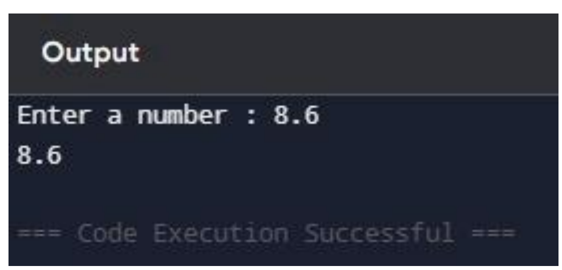
```
#include <iostream> using
namespace std; int main()
{   cout<<"Hello World";

   return 0;
}
```



Q2. WAP to input an integer, decimal and character and display it.

```
#include <iostream> using
namespace std; int main()
{   int a ;
 cout<<"Enter a number : ";
cin>>a;   cout<<a;

   return 0;
}
```


```
#include <iostream> using
namespace std; int main()  {
float a ;
 cout<<"Enter a number : ";
cin>>a;   cout<<a;

   return 0; }
```



```
#include <iostream> using namespace std; int main()  {   char str[20] ;  cout<<"Enter a
character : ";  cin>>str;   cout<<str;

   return 0; }
```

```
Output
Enter a character : Saanvi
Saanvi

=== Code Execution Successful ===
```

Q3. WAP to add, subtract, multiply, divide numbers.

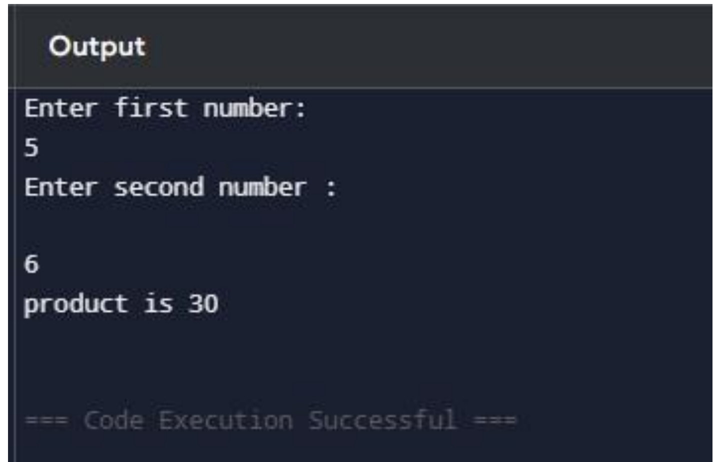
```
#include <iostream> using namespace std; int main()
{  int a;  int b;  int sum;  cout<<"Enter first
number:"<<endl;   cin>>a;  cout<<"Enter second
number :"<<endl;  cin>>b;  sum=a+b;   cout<< "sum
is "<<sum<<endl;
   return 0;
}
```

```
Output
Enter first number:
8
Enter second number :

9
sum is 17

=== Code Execution Successful ===
```

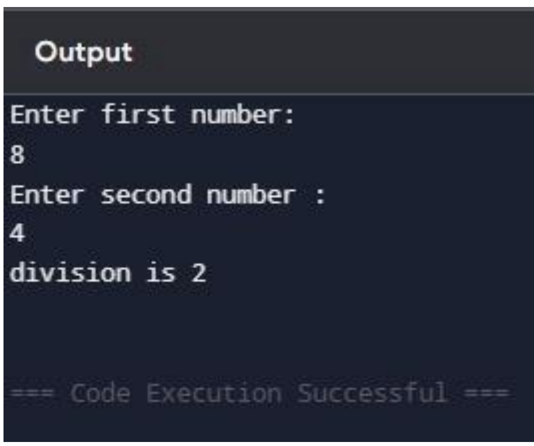
```
#include <iostream> using namespace std; int main()
{  int a;  int b;  int subtract;  cout<<"Enter first
number:"<<endl;   cin>>a;  cout<<"Enter second
number :"<<endl;  cin>>b;  subtract=a-b;   cout<<
"subtract is "<<subtract<<endl;     return 0;
}
```

```
Output
Enter first number:
8
Enter second number :
7
subtract is 1

=== Code Execution Successful ===
```

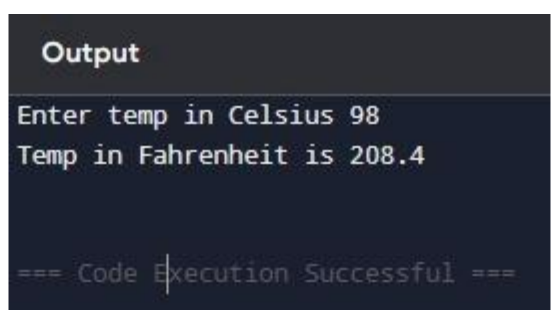
```
#include <iostream> using namespace std; int main()  {  int a;  int b;  int product;  cout<<"Enter
first number:"<<endl;   cin>>a;  cout<<"Enter second number :"<<endl;  cin>>b;  product=a*b;
cout<< "product is "<<product<<endl;
   return 0;
}
```




```
#include <iostream>
using namespace std; int
main()
{  int a;
int b;  int                division;  cout<<"Enter first
                           number:"<<endl;   cin>>a;  cout<<"Enter
second                     number :"<<endl;  cin>>b;  division=a/b;
cout<<                     "division is "<<division<<endl;
   return                  0;
}
```



Q4. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit.[F=9C/5+32]



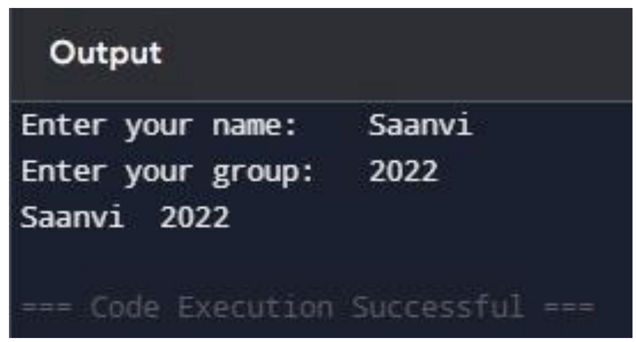

```
#include <iostream> using namespace std; int main()
{  float C;  float F;  cout<<"Enter temp in Celsius ";
cin>>C;
 F=((9*C)/5)+32; cout<<"Temp in
 Fahrenheit is "<<F<<endl;
```

```
    return 0;
}
```

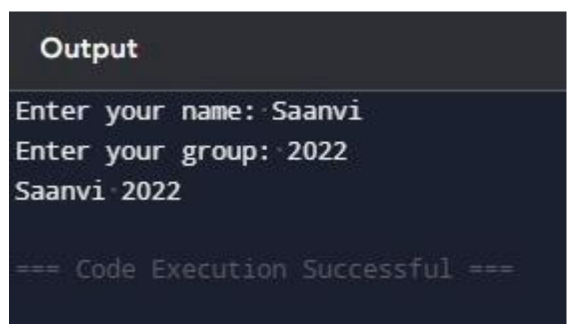
Q5. WAP to format console output using '\n', '\t', '\b' , endl, setw within cout statement

```
#include <iostream> using namespace std; int main()
{     char str[20];     int g;
    cout<<"Enter your name:\n";     cin>>str;
cout<<"Enter your group:\n ";     cin>>g;
cout<<str<<'\n'<<g;     return 0;
}
```


```
#include <iostream> using namespace std; int main()
{     char str[20];     int g;
    cout<<"Enter your name:\t";
cin>>str;     cout<<"Enter your
group:\t";     cin>>g;
cout<<str<<'\t'<<g;
```




```
#include <iostream> #include<iomanip> using
namespace std; int main() {     char str[20];     int g;
    cout<<"Enter your name:\b";
cin>>str;     cout<<"Enter your
group:\b";     cin>>g;
cout<<str<<setw(10)<<g;

    return 0; }
```



return 0;

}

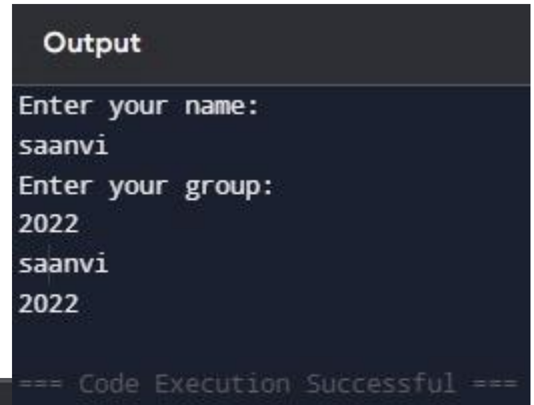
#include <iostream> using namespace std;

int main() { char str[20]; int g;

cout<<"Enter your name:"<<endl; cin>>str;

cout<<"Enter your group:" <<endl; cin>>g;

cout<<str<<endl<<g;



#include

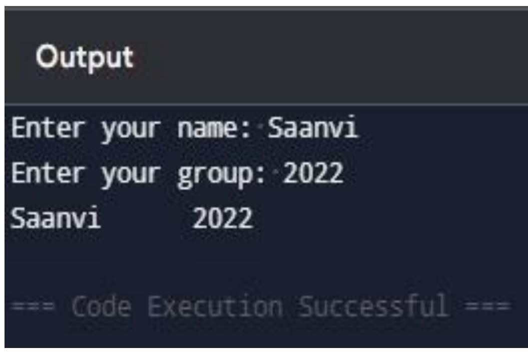
<iostream>

namespace std;

int g;

cout<<"Enter

cout<<"Enter



#include<iomanip> using

int main() { char str[20];

your name:\b"; cin>>str;

your group:\b"; cin>>g;

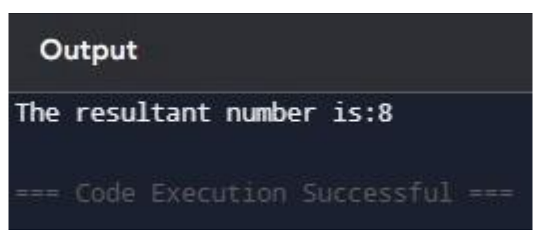
cout<<str<<setw(10)<<g;

return 0;

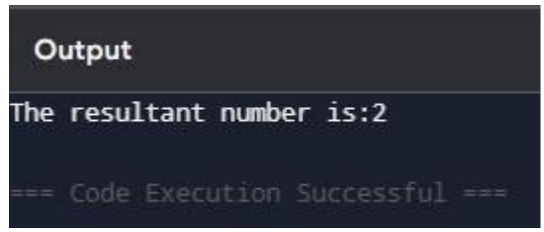
}

Q6. WAP to utilize assignment += and - = type operators.

#include <iostream> using namespace std; int main() {



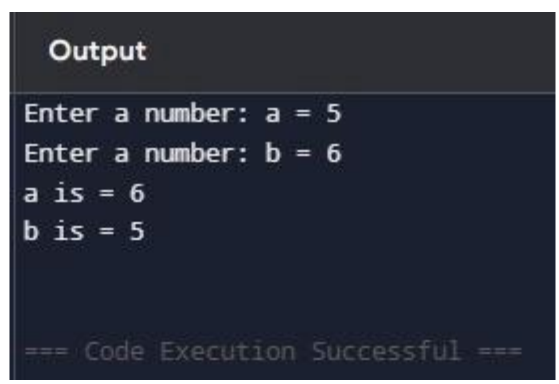
```
    return 0;
}
   int number=5;     number+=3;
cout<<"The resultant number is:",number;
#include <iostream>     using namespace std;
int main() {

        int number=5;        number-=3;        cout<<"The
resultant number is:"<< number;
```




```
return 0;
    }
```

Q7. WAP to swap two numbers using a bitwise operator.

```
#include <iostream>     using namespace std;     int main()      {        int a,b;
        cout<<"Enter a number: a = ";        cin>>a;        cout<<"Enter a number: b = " ;
cin>>b;        a=a^b;        b=a^b;        a=a^b;
```

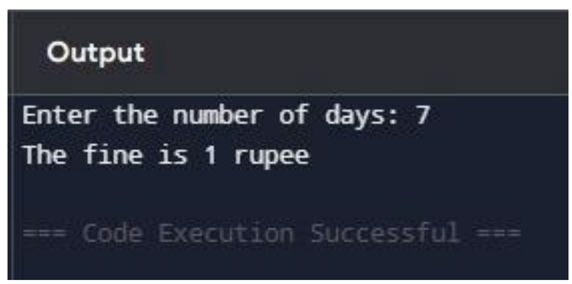


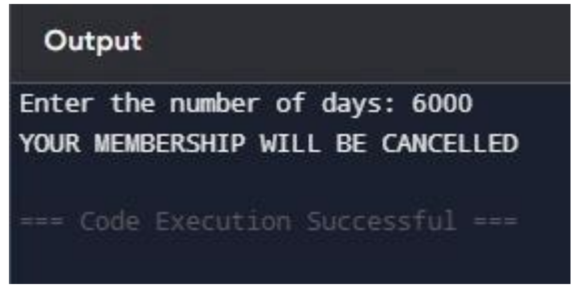
```
    cout<<"a is = "<<a<<endl;
cout<<"b is = "<<b<<endl;

return 0;
  }
```

Q8. Write a program to solve the following problem

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled .WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

```
#include <iostream> using
namespace std;

int main()
{    int days;

  cout << "Enter the number of days: ";     cin >>
days;

  if (days > 0 && days < 6)
  {       cout << "The fine is 50
paise";
  }
  else if (days >= 6 && days < 11)
  {
    cout << "The fine is 1 rupee";
  }
```






```
    else if (days >= 11 && days < 31)
    {
        cout << "The fine is 5 rupees";
    }    else if (days >= 31)
    {
        cout << "YOUR MEMBERSHIP WILL BE CANCELLED";
    }
else
    {
        cout << "Undefined day";
    }

    return 0;
}
```

Q9. WAP to display arithmetic operations using switch-case statement.

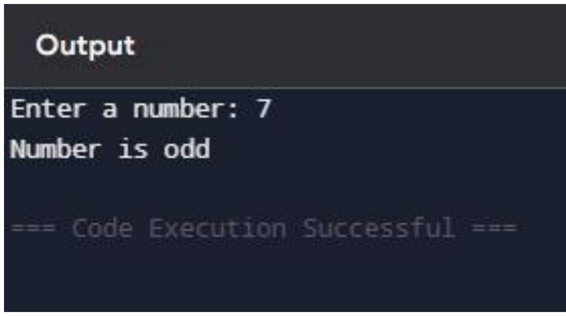
```
#include <iostream> using
namespace std;

int main()
{    int a,
b;    char
op;

    cout << "Enter two numbers: ";
    cin >> a >> b;
```

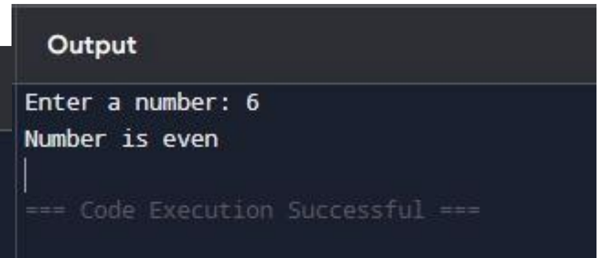
```
    cout << "Enter operator (+, -, *, /): ";     cin >> op;


    switch (op)
    {         case '+':
            cout << "Addition = " << a + b;
break;
         case '-
':
            cout << "Subtraction = " << a - b;
break;
         case
'*':
            cout << "Multiplication = " << a * b;
break;
         case '/':            if (b != 0)
cout << "Division = " << a / b;
            else
                cout << "Division by zero not allowed";
break;


default:
            cout << "Invalid operator";
    }

    return 0;
}
```

```
Output
Enter two numbers: 7 8
Enter operator (+, -, *, /): +
Addition = 15

=== Code Execution Successful ===
```

Q10. WAP to check whether the given number is even or odd (By using if-else and conditional operator).

```
#include <iostream> using namespace std; int main()  {   int a;
```






```
  cout<<"Enter a number: ";
cin>>a;
  if(a%2==0)
  {
    cout<<"Number is even ";
  }

  else
  cout<<"Number is odd";

    return 0;
}
```